



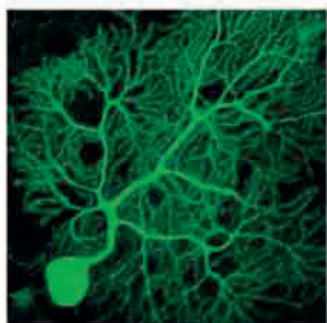
Journalists in Hong Kong wear face masks to reduce the risk of airborne infection by the H5N1 virus, commonly known as bird flu.

6 THE NUMBER OF FATALITIES IN THE 18 PEOPLE INFECTED WITH BIRD FLU IN 1997

CHESS-PLAYING COMPUTER PROGRAMS first appeared in the late 1950s. Increases in computing power led to more powerful programs. In 1997, for the first time, a **computer won a match** against the reigning world chess champion. **IBM's Deep Blue** computer won two of the six games of the match, with three draws, while its human opponent, Russian **grandmaster Garry Kasparov**, won only one game.

The popular Internet search engine **Google**—originally called BackRub—**got its name** this year. This new name was derived from the word googol—a mathematical term for the number represented by 1 followed by one hundred zeroes. The creators of Google—American computer scientist **Larry Page** (b.1973) and Russian-born computer scientist **Sergey Brin** (b.1973)—were still developing the search engine at Stanford University, California. They incorporated the company, Google Inc., in 1998.

A particularly **virulent influenza virus** known as H5N1, which had affected birds since the 1950s, **crossed the species barrier** into humans. An outbreak of the disease it causes, nicknamed bird flu, killed six people in Hong Kong. Health authorities were concerned that the disease could become a pandemic and issued hygiene advice to international



Green fluorescent protein
A micrograph shows a cell taken from a mouse's brain. The cell is producing green fluorescent protein (GFP), a substance used to track gene expression.

travelers and to people working in the poultry trade. Although **other bird flu outbreaks** have occurred since then, fears of a pandemic have not been realized.

At a conference in December in Kyoto, Japan, the United Nations reached an **agreement called the Kyoto Protocol** that relates to the UN Framework Convention on Climate Change (UNFCCC, see 1992). The countries that signed and ratified the agreement were committed to reducing

Kyoto Protocol emission targets
Emissions targets are shown for the first assessment period (2008–12) of the Kyoto Protocol. Most countries had to reduce their emissions by the end of this period; some had leeway to increase them. The US signed the protocol but did not ratify it.

their emissions of greenhouse gases—most importantly, carbon dioxide released by burning fossil fuels (see pp.326–27). Each participating nation had a target: its emissions for the period 2008–12 had to be reduced by a certain percentage compared with the emissions in a base year (in most cases, 1990). The targets did not take into account emissions from aviation and international shipping. At a conference at Doha, Qatar, in December 2012, parties to the **UNFCCC agreed to new targets** for the second assessment period—2013 to 2020.

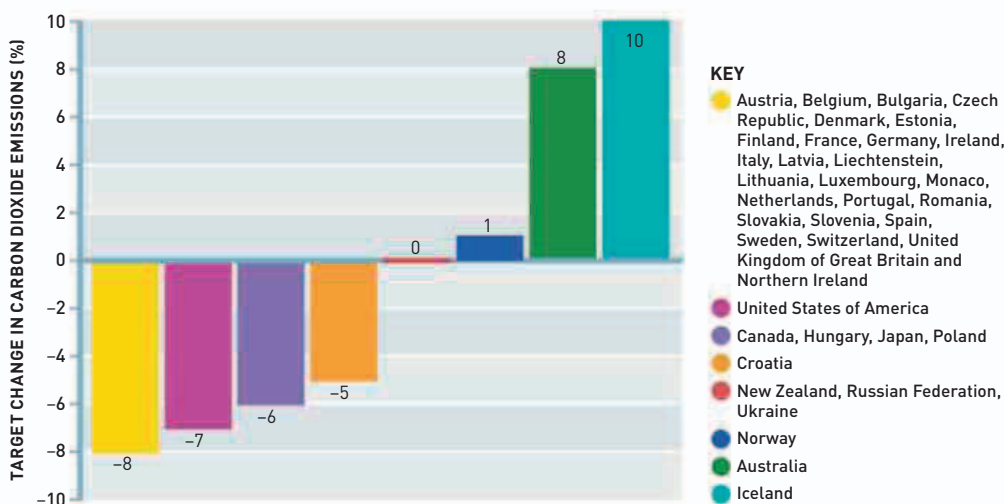
A team led by Japanese geneticist **Masaru Okabe** hit the headlines when they produced **genetically modified mice** that

“DEEP BLUE WAS ONLY INTELLIGENT THE WAY YOUR PROGRAMMABLE ALARM CLOCK IS INTELLIGENT. NOT THAT LOSING TO A \$10 MILLION ALARM CLOCK MADE ME FEEL ANY BETTER.”

Garry Kasparov, Russian chess grandmaster, 1997

produce a green glow under ultraviolet light. The glow is produced by a compound called **green fluorescent protein (GFP)**, which exists naturally in certain jellyfish. The gene that codes for GFP was first sequenced in 1994, and the protein is now an important tool in molecular biology. By inserting the GFP-coding gene from the jellyfish

into a part of the genomes of other organisms, researchers can tell when and whether that part of the **genome is being activated**. Okabe injected the gene into mouse embryos with the hope of tracking the development of the mice's sperm cells, but instead, the protein was produced in nearly every type of cell in the mice's bodies.



March Standard published for cable modems that give broadband Internet access

May 11 IBM's Deep Blue became the first computer to beat world champion chess player

May Glowing mice—whose cells produced green fluorescent protein—bred for first time

June Wireless networking protocol 802.11, dubbed Wi-Fi, released

October 9 Warnings of a possible flu pandemic after the first cases of humans infected with bird flu

September 15 Google Inc. registers the domain name google.com

December 11 The Kyoto Protocol adopted at the Kyoto global warming conference